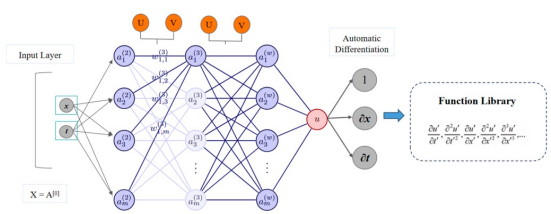
Initial Results

Motivation

1. Recover an interpretable evolution law from spatiotemporal data
2. A method that not only fitting black-box predictor

SG Smoothing
A way to **smooth noisy data** without destroying important features

PySR
A Python Library for **symbolic regression**



(B)

Workflow

Generate Clean Training Data

↓

Corrupt field with synthetic noise

↓

Build strong/weak-form regression problem

↓

Strong Form (SG Smoothing Required) | Weak Form (No Smoothing Required)

↓

Candidate library (hand-chosen per PDE)

↓

Sparse regression via STLSQ or PySR

↓

Compare coefficients to ground truth

↓

Re-simulate those rediscovered GEs

↓

Results (Tables + Graphs)

Governing Equations

Burgers Equation: $u_t = -u\,u_x + 0.1\,u_{xx}$

KdV Equation: $u_t = -u\,u_x - 0.0025\,u_{xxx}$

Klein-Gordon Equation: $u_{tt} = 0.5\,u_{xx} - 5\,u$

Convection-Diffusion: $u_t = -u_x + 0.25\,u_{xx}$

Chaffee-Infante: $u_t = u_{xx} - u + u^3$

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Results

Table 3: Burgers coefficient recovery. True: $\xi_{u*u_x} = -1.0$, $\xi_{u_{xx}} = +0.1$.

	SG + PySR			Weak Form			
Noise	$\hat{\xi}_{u*u_x}$	$\hat{\xi}_{u_{xx}}$	Error	$\hat{\xi}_{u*u_x}$	$\hat{\xi}_{u_{xx}}$	WF R^2	Error
0%	−0.9898	+0.107	0.039	−1.0000	+0.1000	1.0000	0.000
20%	−0.9494	+0.051	0.269	−0.9999	+0.1004	0.9997	0.002
40%	−0.7624	+0.019	0.522	−0.9997	+0.1009	0.9987	0.005
60%	−0.5712	+0.008	0.677	−0.9992	+0.1014	0.9971	0.007
80%	−0.5163	−0.006	0.769	−0.9986	+0.1020	0.9948	0.011
100%	−0.3737	−0.005	0.837	−0.9977	+0.1027	0.9918	0.015

Improvement factor at 60% noise: 0.677/0.007 ≈ **97**×

Table 4: KdV coefficient recovery. True: $\xi_{u*u_x} = -1.0$, $\xi_{u_{xxx}} = -0.0025$. The SG pipeline cannot recover u_{xxx} at any noise level; the weak form does so to four significant figures.

	SG + PySR			Weak Form			
Noise	$\hat{\xi}_{u*u_x}$	$\hat{\xi}_{u_{xxx}}$	Error	$\hat{\xi}_{u*u_x}$	$\hat{\xi}_{u_{xxx}}$	WF R^2	Error
0%	−0.4461	−0.00138	0.501	−1.0000	−0.002500	1.0000	0.0001
1%	−0.4391	−0.00129	0.523	−1.0001	−0.002503	1.0000	0.0001
5%	−0.4380	−0.00139	0.494	−1.0002	−0.002504	0.9999	0.0006
10%	−0.3831	−0.00100	0.608	−1.0000	−0.002494	0.9995	0.0013
20%	−0.2725	−0.00061	0.741	−0.9990	−0.002483	0.9981	0.0036
30%	−0.2200	−0.00042	0.807	−0.9970	−0.002469	0.9957	0.0071

Improvement factor at 10% noise: 0.608/0.0013 ≈ **580**×

Table 5: Klein-Gordon coefficient recovery. True: $\xi_{u_{tt}} = +0.5$, $\xi_u = -5.0$. The weak form performs well at 0% noise but degrades sharply above 10%; the SG pipeline is consistently poor.

	SG + PySR			Weak Form			
Noise	$\hat{\xi}_{u_{tt}}$	$\hat{\xi}_u$	Error	$\hat{\xi}_{u_{tt}}$	$\hat{\xi}_u$	WF R^2	Error
0%	+0.0166	−9.62	0.946	+0.4965	−5.037	1.0000	0.007
10%	+0.0993	−9.24	0.825	+0.135	−8.579	0.9999	0.723
20%	+0.176	−8.78	0.702	+0.046	−9.461	0.9998	0.901
30%	+0.209	−7.91	0.582	+0.021	−9.709	0.9995	0.950
40%	+0.228	−8.71	0.642	+0.011	−9.819	0.9992	0.971
60%	+0.235	−9.30	0.695	0.000	−9.953	0.9982	0.995
80%	+0.233	−12.05	0.973	0.000	−9.977	0.9968	0.998

Table 6: Convection-Diffusion coefficient recovery. True: $\xi_{u_x} = -1.0$, $\xi_{u_{xx}} = +0.25$. The weak form maintains sub-1% error up to 80% noise; the SG pipeline degrades rapidly.

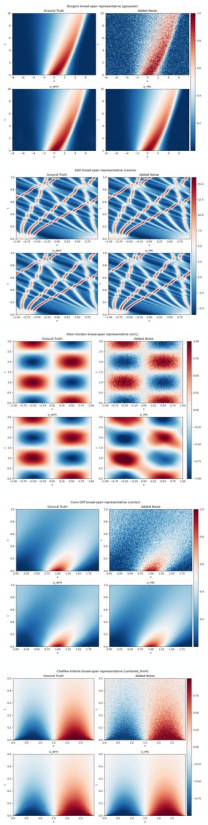
	SG + PySR			Weak Form			
Noise	$\hat{\xi}_{u_x}$	$\hat{\xi}_{u_{xx}}$	Error	$\hat{\xi}_{u_x}$	$\hat{\xi}_{u_{xx}}$	WF R^2	Error
0%	−0.866	+0.109	0.349	−1.000	+0.250	1.0000	0.000
20%	−0.692	+0.009	0.637	−1.001	+0.251	0.9989	0.002
40%	−0.499	+0.003	0.745	−1.002	+0.251	0.9957	0.003
60%	−0.303	+0.001	0.846	−1.003	+0.250	0.9902	0.002
80%	−0.185	+0.001	0.906	−1.003	+0.250	0.9826	0.002
100%	−0.140	+0.000	0.929	−1.003	+0.248	0.9729	0.005
120%	−0.118	+0.001	0.940	−1.002	+0.246	0.9610	0.008

Improvement factor at 60% noise: 0.846/0.002 ≈ **430**×

Table 7: Chaffee-Infante coefficient recovery. True: $\xi_{u_{xx}} = +1.0$, $\xi_u = -1.0$, $\xi_{u^3} = +1.0$. Both pipelines struggle; the weak form reduces error by $\sim 17\times$ but collinearity prevents accurate recovery.

	SG + PySR				Weak Form			
Noise	$\hat{\xi}_{u_{xx}}$	$\hat{\xi}_u$	$\hat{\xi}_{u^3}$	Err	$\hat{\xi}_{u_{xx}}$	$\hat{\xi}_u$	$\hat{\xi}_{u^3}$	WF R^2	Err
0%	−0.002	−13.3	+11.3	7.88	+0.878	−2.73	+2.65	0.999	1.17
20%	−0.004	−12.5	+10.5	7.34	+0.936	−1.87	+1.76	0.989	0.57
40%	+0.004	−12.4	+10.2	7.18	+0.878	−1.82	+1.30	0.959	0.41
60%	+0.004	−11.1	+8.86	6.32	+0.768	−2.04	+0.96	0.914	0.44
80%	+0.002	−11.3	+8.48	6.26	+0.656	−2.10	+0.48	0.856	0.65

Improvement factor at 40% noise: 7.18/0.41 ≈ **17**×



CS598-SML

[1]
S. Zhang, Y. Huang, N. Yi, and shihan Zhang, "Robust PDE discovery under sparse and highly noisy conditions via attention neural networks," Jun. 22, 2025, *arXiv*: arXiv:2506.17908. doi: 10.48550/arXiv.2506.17908.